Sachin Ashok

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LinkedIn Google Scholar GitHub

Research Interests

Broadly interested in networked systems with recent emphasis on enhancing performance, monitoring, and root cause analysis for microservices and cloud systems.

Education

2021 - **PhD.** in Computer Science; CGPA: 4.0
University of Illinois at Urbana-Champaign
Advisors: **Prof. Radhika Mittal**, **Prof. Philip Brighten Godfrey**

2014 - 2018 **B.Tech.** in Computer Science and Engineering
National Institute of Technology, Trichy

Publications

Murphy: Performance Diagnosis of Distributed Cloud Applications
Vipul Harsh, Wenxuan Zhou, **Sachin Ashok**, Radhika N. Mysore, Brighten Godfrey, Sujata Banerjee
In Proceedings of ACM SIGCOMM, 2023.

Fast and Efficient Look-Ups via Data-Driven FIB Designs
Sachin Ashok*, Aditi Partap*, Ammar Tahir* (* = equal contribution)
In Proceedings of ACM SIGCOMM FIRA Workshop, 2022.

Data-Driven Network Path Simulation with iBox
Sachin Ashok, Shubham Tiwari, Nagarajan Natarajan, Venkata N. Padmanabhan, Sundararajan Sellamanickam
In Proceedings of ACM SIGMETRICS, 2022.

Leveraging Service Meshes as a New Network Layer
Sachin Ashok, P. Brighten Godfrey, Radhika Mittal
In Proceedings of ACM HotNets, 2021.

iBox: Internet in a Box
Sachin Ashok, Sai Surya Duvvuri, Nagarajan Natarajan, Venkata N. Padmanabhan, Sundararajan Sellamanickam, Johannes Gehrke
In Proceedings of ACM HotNets, 2020.

Reinforcement Learning for Bandwidth Estimation and Congestion Control in Real-Time Communications
Joyce Fang, Martin Ellis, Bin Li, Siyao Liu, Yasaman Hosseinkashi, Michael Revow, Albert Sadovnikov, Ziyuan Liu, Peng Cheng, **Sachin Ashok**, David Zhao, Ross Cutler, Yan Lu, Johannes Gehrke
In Proceedings of NeurIPS MLForSystems workshop, 2019.

Research Experience

2018 - 2020 **Microsoft Research, India** **Research Fellow**
Mentors: **Venkat Padmanabhan**, **Naga Natarajan**, **Sundar Sellamanickam**, **Johannes Gehrke**

- **iBox: Internet in a Box**: Designed a data-informed network simulator for recreating the behaviour of paths in a target network such as the Internet.
  Published at SIGMETRICS'22.

- **R3Net: Reinforcement Learning for Bandwidth Control**: Developed realistic network environments to train RL agents for bandwidth estimation & congestion control. Published at MLForSystems'19.

Industry Experience

2016 - 2017 **Mozilla Organization** **Intern**
Mentors: **Franziskus Kiefer**, **Tim Taubert**

- **Blake2 support for Mozilla's NSS library**: Implemented a Blake2 (a cryptographic hash function) module for Mozilla's Network Security Services (NSS) library. The library is used by Google, RedHat, and Mozilla products like Thunderbird.

2017 **Samsung R&D, India** **Intern**
Mentors: **Pratibha Moogi**, **Karthikeyan Somanathan**

- **Object localization framework**: Developed an end-to-end system to leverage deep learning models using custom convolutional neural networks for localization of objects in a given image.

Invited Talks

- Distributed Tracing without the Pain!
  KubeCon 2022, Detroit, USA
- Fast and Efficient Lookups via Data-Driven FIB Designs (presented virtually)
  FIRA@SIGCOMM 2022, Amsterdam, Netherlands
- Data-Driven Network Path Simulation with iBox
  ACM SIGMETRICS 2022, Indian Institute of Technology Bombay (IIT-B), India
- Leveraging Service Meshes as a New Network Layer (presented virtually)
  CSL Student Conference 2022, University of Illinois at Urbana-Champaign
- Leveraging Service Meshes as a New Network Layer (presented virtually)
  HotNets 2021, University of Cambridge, Virtual Event
- iBox: Internet in a Box (presented virtually)
  HotNets 2020, University of Chicago, Virtual Event
- iBox: Internet in a Box for Realistic Network Simulation
  Bandwidth Control Workshop 2019, Microsoft Research, Redmond

Honors

- Invited for demo at **Microsoft TechFest**, 2020.
- Finalist, **Pragyan CTF**, an international level security contest held by **NIT Trichy**, 2018.
- Among top 100 chosen by the **Indian Institute of Science (IISc)** for **CSA Undergraduate Summer School**, 2017.
- Among top 25 chosen by **Institute of Mathematical Sciences (IMSc)** for **Summer Research Program**, 2017.
- Among 45 chosen worldwide for **Mozilla Winter of Security**, 2016.

Skills

Languages: C++, C, Python, Bash
Tools: Kubernetes, eBPF, Protobuf, ns-2/3, gRPC